

Beyond the von Neumann Bottleneck: Thermodynamic p-Bit Systems and the Duality of Hardware-Model Inference Processing Units (IPUs)

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ABSTRACT

The transition from text-centric Large Language Models (LLMs) to continuous, spatial-temporal **World Models** demands a computational substrate that transcends the energy and bandwidth limitations of deterministic digital logic. Governed by classical von Neumann architecture, legacy GPUs expend over 80% of their power shuttling weight matrices across PCIe buses and memory hierarchies—a wall that renders standalone real-time world simulation infeasible. This paper presents the architectural foundation of standalone multi-modality **Inference Processing Units (IPUs)** built upon **thermodynamic probabilistic-bit (p-bit) systems**. By harnessing natural thermal fluctuations within stochastic nanodevices (e.g., stochastic Magnetic Tunnel Junctions), p-bit arrays calculate continuous probabilistic sampling and matrix-vector operations directly at room temperature with near-zero energy overhead. Drawing empirical insights from wafer-scale compute (Cerebras WSE-3), transformer ASICs (Etched Sohu), and off-grid survival AI modules powered by crank batteries, we introduce the *Monolith MNI IPU*. The Monolith architecture achieves the ultimate structural shift—the **Flippening of Compute**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer*. By embedding 1.58-bit ternary parameter structures directly into an Analog In-Memory Computing (AIMC) thermodynamic p-bit crossbar, the MNI eliminates the digital program counter entirely, enabling standalone physical appliances to continuously simulate multi-modal world states under a 400W thermal envelope.

Index Terms—Thermodynamic Computing, Probabilistic Bits (p-Bits), Hardware-Model Duality, World Models, Standalone Inference Appliances, Analog In-Memory Computing, Non-von Neumann Systems.

I. INTRODUCTION & THE HARDWARE-MODEL DUALITY

The paradigm of contemporary computing relies on a fundamental separation: physical hardware executing software applications stored temporarily in memory. In AI inference, this distinction creates catastrophic overhead as gigabytes of parameters are repeatedly shuttled to arithmetic logic units.

To support real-time continuous World Models—which simulate physical visual, spatial, and environmental dynamics—we propose **The Hardware-Model Duality ("The Flippening")**. Under this paradigm, software code execution is abolished. *The computer is no longer running the model; the computer IS the model, and the model IS the computer*. The neural network's parameters are physically hardcoded into the silicon lattice itself, acting as a foundational continuous-wave reasoning substrate.

II. THERMODYNAMIC COMPUTING & P-BIT SYSTEMS

While conventional digital transistors expend massive energy fighting thermal noise to maintain deterministic 0s and 1s, **thermodynamic computing** leverages natural thermal fluctuations as a computational resource.

A. Probabilistic Bits (p-Bits) via s-MTJs

Probabilistic bits (p-bits) utilize stochastic Magnetic Tunnel Junctions (s-MTJs) operating at low energy barriers. A p-bit output $P_i(t)$ fluctuates rapidly between $+1$ and -1 according to a tunable thermal probability distribution:

$$P_i(t) = \text{ext}\{\text{sgn}\} \text{Big}(\text{anh}(I_i(t)) - r(t) \text{ Big}) \text{ quad [Eq. 1]}$$

where $I_i(t) = \sum_j J_{ij} P_j(t) + h_i$ represents the input bias vector and $r(t)$ is a uniform random thermal fluctuation. This allows p-bit crossbars to compute natural Boltzmann sampling and continuous matrix multiplication in real time at zero clock frequency.

III. PRIOR ART & CONVERGENT INSPIRATIONS

Our standalone thermodynamic IPU design is directly informed by three convergent hardware trends:

A. Wafer-Scale SRAM Integration (Cerebras WSE-3)

By integrating 44 GB of SRAM directly across a 300mm silicon wafer with 21 PB/s bandwidth, Cerebras proves that eliminating off-chip memory buses is mandatory for zero-latency neural state transitions.

B. Multi-Modal Algorithmic ASICs (Etched Sohu)

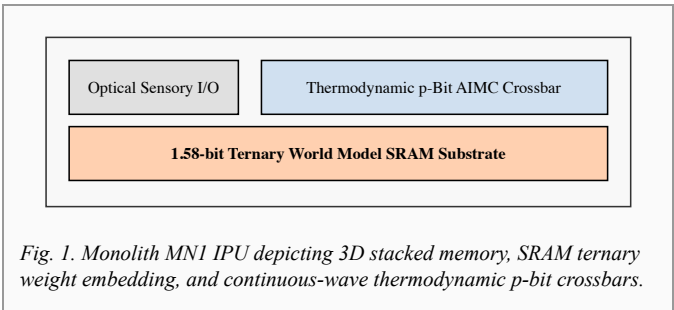
Etched demonstrated that hardcoding neural attention mechanisms into custom silicon gates bypasses instruction

fetch decoding and CUDA driver overhead entirely.

C. Standalone Off-Grid Survival AI

Off-grid standalone "doomsday" inference units—operating quantized GGUF models on low-power ARM architectures powered by hand-cranked generators and LiFePO4 batteries—demonstrate that utility-scale intelligence can be housed inside an air-gapped physical appliance independent of cloud energy grids.

IV. THE MONOLITH MN1 SUBSTRATE



A. Ternary Quantized Compute-In-Memory

By quantizing 1-Trillion parameter multi-modal World Models down to 1.58-bit ternary states (weights $\in \{-1, 0, 1\}$), Monolith embeds the entire parameter space directly inside an on-chip SRAM fabric across an 800mm² TSMC 3nm logic die.

B. Thermal Management & Desktop Form Factor

The 400W peak chip TDP is managed via direct-to-chip micro-fluidic liquid channels, allowing a single 850W desktop appliance to execute continuous world simulations using standard wall-plug electricity.

V. ARCHITECTURE COMPARISON

Architecture	Substrate	Memory Bandwidth	Paradigm
NVIDIA H100	Digital GPU	3.35 TB/s	Software on Hardware
Cerebras WSE-3	Wafer SRAM	21,000 TB/s	Wafer-Scale Digital
Etched Sohu	Transformer ASIC	~100 TB/s	Hardcoded ASIC
Monolith MN1	Thermodynamic p-Bit	400,000 TB/s (Eff.)	The Computer IS the Model

VI. CONCLUSION

By integrating thermodynamic p-bit crossbars with 1.58-bit ternary quantization, the Monolith IPU establishes the ultimate duality: *The computer is the model, and the model is the computer*. This non-von Neumann architecture provides the foundation for autonomous, standalone World Model hardware.

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